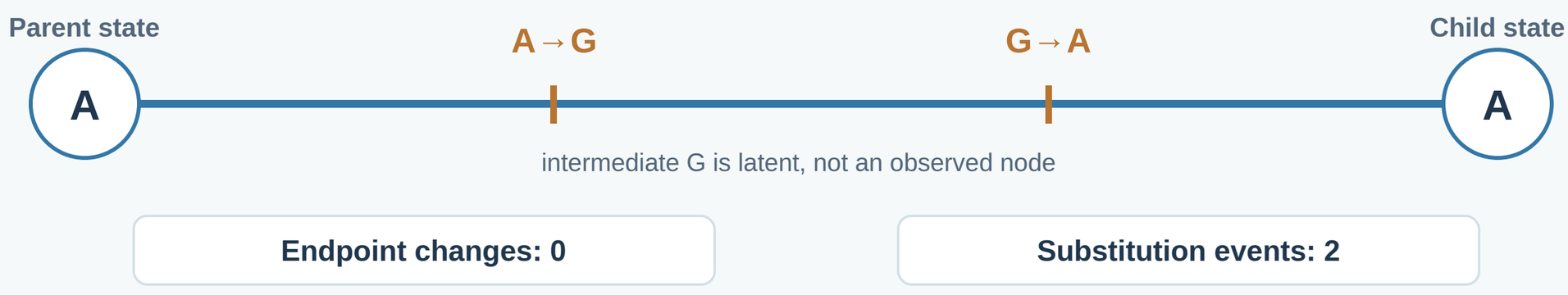
S2 Shared inputs do not imply shared inference targets

A single-branch substitution history illustrates a difference in the quantities methods estimate

CONCEPTUAL EXAMPLE · ONE PHYLOGENETIC BRANCH



DISTINCT ANALYTICAL ENDPOINTS

Analytical output	Question addressed	Appropriate endpoint
Ancestral state	Nucleotide at an internal phylogenetic node	Node-state agreement
Branch endpoint	Parent-to-child state difference	Net change or state-pair agreement
Substitution history	Latent events along an evolutionary branch	Event counts / mapped histories
Homoplasy	Repeated or convergent changes across branches	Recurrence-specific metrics

Benchmark eligibility also requires compatible rooted trees, branch identities, site coordinates, substitution models and missing-data rules.

Illustration is not an observed result and does not establish native JC69 compatibility for any new software.